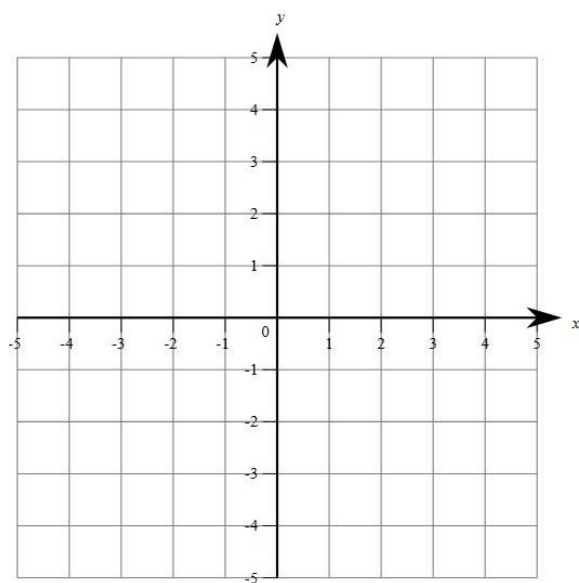


All electronic devices and notes are prohibited. All work must be shown for credit.

1) Given $P(x) = -2x^2 - 12x - 15$.

(a) Write the function in vertex form.

(b) Graph using the vertex, intercepts and axis of symmetry. Be accurate with all integer ordered pairs.



(c) State the domain and range.

2) Find the quadratic function whose vertex is at $(2, -3)$ and passes through the point $(0, 5)$.

3) Two chicken coops are to be built adjacent to one another using 120 ft of fencing.

(a) Find a function that models the enclosed area of the 2 pens. Draw a picture and label with the variables that you use.

(b) Find the dimensions that yield the maximum enclosed area.

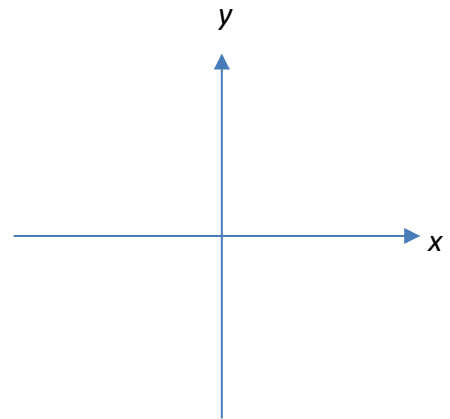
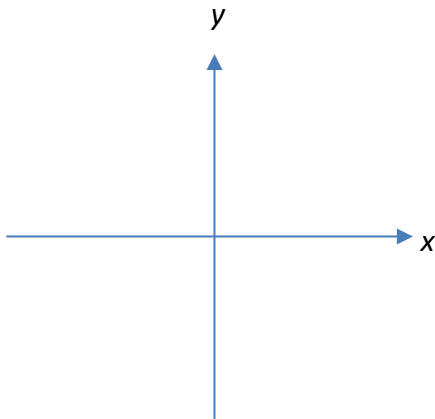
4) Given $f(x) = -\frac{1}{4}(x+1)^3 + 2$ and $r(x) = 2 - \frac{1}{(x+3)^2}$.

(a) Determine the intercepts.

(b) Graph. Clearly indicate any important points and draw in and label any asymptotes.

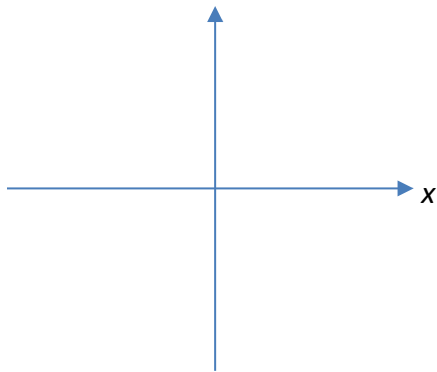
$$f(x) = -\frac{1}{4}(x+1)^3 + 2$$

$$r(x) = 2 - \frac{1}{(x+3)^2}$$

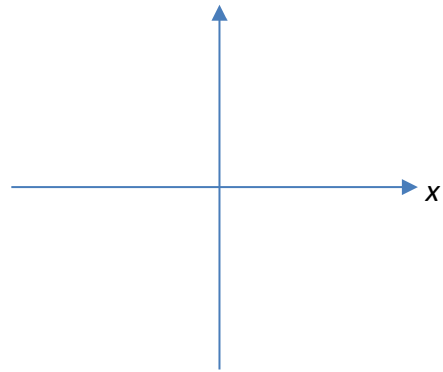


6) Graph the following polynomials.

(a) 6th degree: 4 x-intercepts



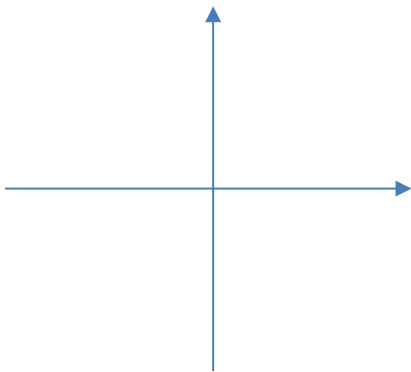
(b) -5th degree: 1 x-intercept



7) Given $p(x) = -6x^3 - 9x^2 + 60x$ and $m(x) = x^5 - 10x^4 + 25x^3$

(a) Completely factor each function.

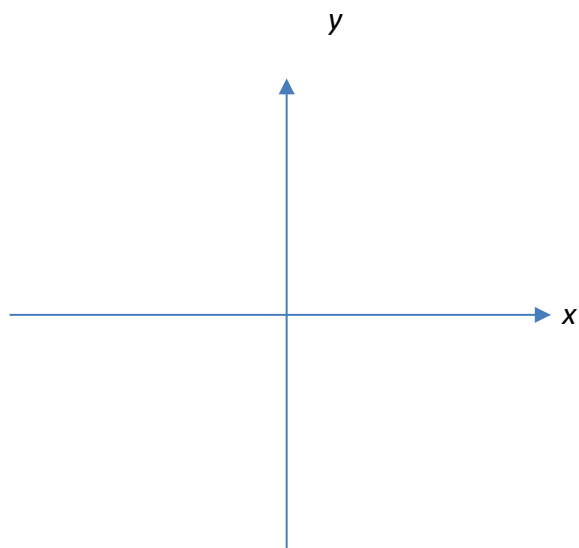
(b) Graph by analyze turning points, intercepts, and end behavior. Mark all intercepts as numbers on the axes.



8) For the function $f(x) = \frac{x^3 - 3x^2 - 4x}{x+1}$

(a) Write the function in factored, reduced form.

(b) Graph and determine domain, intercepts, asymptotes and holes (deleted points) that exist.



Unit 9 Review

1. Let $f(x) = \frac{2x+1}{3-x}$. What is the range of $f^{-1}(x)$? Write your answer in interval notation.

2. What is the domain of $h(x) = 4 - \log(x + 4)$?

3. Simplify

a. $\log_2 16$

b. $\log_2 \left(\frac{1}{16}\right)$

c. $\log_{\frac{1}{2}} 16$

d. $\log_5 \sqrt[3]{5^4}$

4. Graph $f(x) = e^x$ and its inverse function on the same set of axes. What is the inverse function? Label all important aspects of both functions.

5. Find the x-intercept of each function.

a. $\frac{1}{2} \log_2(3 - x) + 2$

b. $g(x) = 2e^{x+1} - 3$

6. Write $\frac{1}{2} \log 16x^2 - \log x + \log(x-1)$ with 1 log and a coefficient of 1.

7. Identify all asymptotes, intercepts, and deleted points for $f(x) = \frac{(x+1)(x+2)(x-3)}{x^2-1}$. Please give your answers in the appropriate form (points and lines) and label their role in the graph (ie vertical asymptote,...)

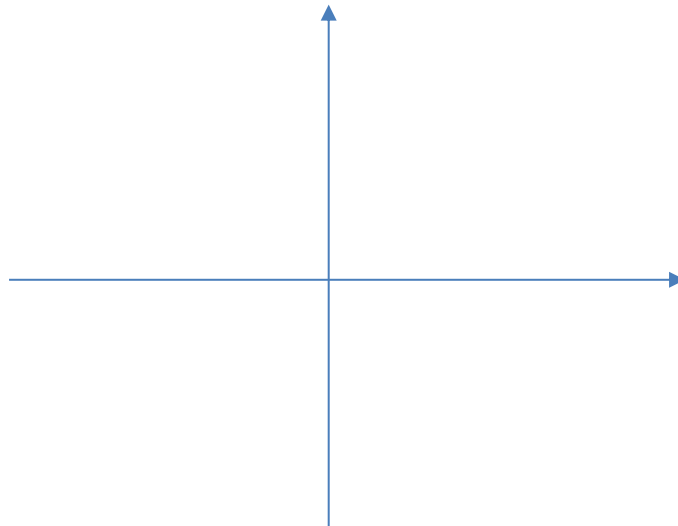
VA(s): _____

x-int: _____

HA(s): _____

y-int: _____

DP: _____



x-intervals:

Above or
Below:

Justification:

8.Solve

a. $4^{x-3} = \frac{8}{2^{x-1}}$

b. $3^{x+5} - 6 = 10$

c. $\frac{1}{2} \log_{16} x = \frac{1}{8}$

d. $\log(x-1) - 2\log 3 = \log(6-x)$

9. The function $g(x) = \left(\frac{1}{4}\right)^{x-2} + 1$ is a transformation of a basic function.

a. What is the basic function?

b. Give the transformations applied to the basic function to get to $g(x)$, in order.

c. Graph $g(x)$.